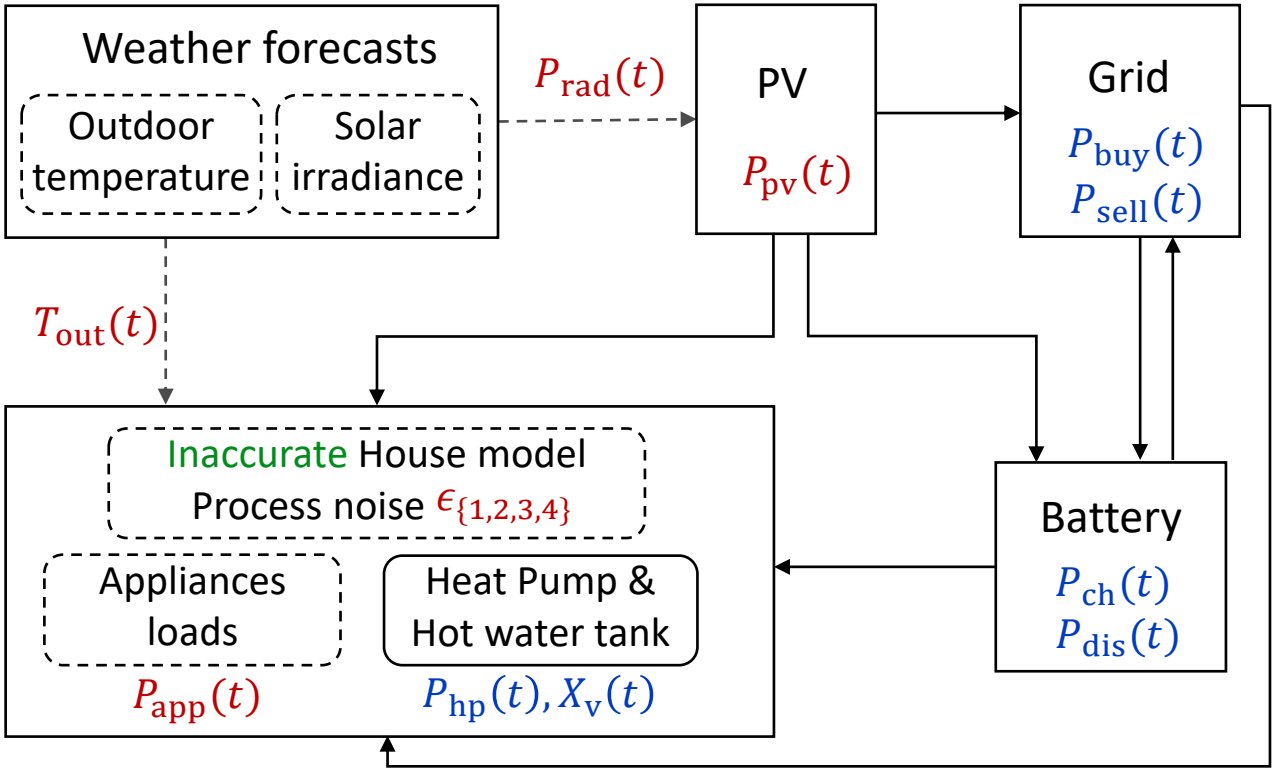

Weather forecasts
Outdoor temperature
Solar irradiance
$P_{\mathrm{rad}}(t)$
PV
$P_{\mathrm{pv}}(t)$
Grid
$P_{\mathrm{buy}}(t)$
$P_{\mathrm{sell}}(t)$
$T_{\mathrm{out}}(t)$
Inaccurate House model
Process noise $\epsilon_{\{1,2,3,4\}}$
Appliances loads
$P_{\mathrm{app}}(t)$
Heat Pump & Hot water tank
$P_{\mathrm{hp}}(t), X_{\mathrm{v}}(t)$
Battery
$P_{\mathrm{ch}}(t)$
$P_{\mathrm{dis}}(t)$